

AP Statistics MCQ Practice Exam 10

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PART A, QUESTIONS

APSTAT-MCQ10-001

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A frequency table for the number of reusable containers returned in one visit shows 1 container for 4 visits, 2 containers for 3 visits, and 5 containers for 1 visit. What are the mean and median numbers returned?

- A) Mean = 1.875 and median = 1.5
- B) Mean = 1.5 and median = 1.875
- C) Mean = 2.0 and median = 2.0
- D) Mean = 2.5 and median = 1.5

APSTAT-MCQ10-002

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

Separate 95% confidence intervals for the germination proportions of seed varieties A and B overlap slightly. A researcher concludes that a two-sided test of $H_0: p_A = p_B$ must therefore fail to reject at $\alpha = 0.05$. Which assessment is correct?

- A) The conclusion is always correct because any interval overlap proves equality.
- B) The conclusion is correct only because both intervals use 95% confidence.
- C) The conclusion is false because any amount of overlap proves the proportions differ.
- D) Overlap of two separate intervals does not by itself determine the two-proportion test; the standard error of $\hat{p}_A - \hat{p}_B$ must be analyzed directly.

APSTAT-MCQ10-003

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Events A and B satisfy $P(A) = 0.70$ and $P(B) = 0.60$. What is the smallest possible value of $P(A \cap B)$?

- A) 0
- B) 0.30
- C) 0.42
- D) 0.60

APSTAT-MCQ10-004

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Formulate Questions

Prompt:

Independent random samples of production batches will be used to compare fermentation times for recipe A and recipe B. Which parameter matches the goal of estimating the difference in population mean fermentation time, A minus B?

- A) $\mu_A - \mu_B$
- B) $\bar{x}_A - \bar{x}_B$
- C) $p_A - p_B$
- D) μ_d for paired within-batch differences

APSTAT-MCQ10-005

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

A scatterplot has a strong positive linear pattern. A new observation is added with x near $\bar{x}$ but y far above the existing pattern. Which effect is most plausible?

- A) It increases squared residual variation and weakens $|r|$, but its x -position gives it relatively little leverage on the slope.
- B) It has high leverage because every vertical outlier must have an extreme x -value.
- C) It strengthens the correlation because its y -value is large.
- D) It leaves the fitted line and correlation unchanged because x equals approximately $\bar{x}$.

APSTAT-MCQ10-006

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A statewide professional association wants to estimate members' satisfaction, but previous surveys had substantial nonresponse. Which plan best preserves probability sampling while addressing nonresponse?

- A) Post an unrestricted survey link and keep it open until the desired sample size is reached.
- B) Replace each nonrespondent with the first volunteer from the same city.
- C) Draw an SRS from the membership roster, contact selected members repeatedly through several modes, and document response patterns.
- D) Survey only members attending the annual conference because they are easy to locate.

APSTAT-MCQ10-007

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

In a comparison of two population proportions, sample 1 has 54 successes among 90 observations and sample 2 has 30 successes among 50 observations. For testing $H_0: p_1 = p_2$ against a two-sided alternative, what are the z statistic and p-value?

- A) $z = 1$ and $p\text{-value} \approx 0.317$
- B) $z = 0$ and $p\text{-value} = 1$
- C) $z = -1$ and $p\text{-value} \approx 0.317$
- D) The statistic is undefined because the sample sizes differ.

APSTAT-MCQ10-008

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

For deliveries from a distribution center, $P(\text{late} \mid \text{rain}) = 0.12$ and $P(\text{late}) = 0.12$, with $P(\text{rain}) > 0$. Which statement follows?

- A) Rain and lateness are mutually exclusive.

- B) Rain causes exactly 12% of deliveries to be late.
- C) Every rainy day has exactly the same number of late deliveries.
- D) Rain and lateness are independent in the probability model because conditioning on rain does not change the late-delivery probability.

APSTAT-MCQ10-009

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

For a difference in population means, software reports an estimate of 7.2 units and $t = 2.40$ for testing a null difference of 0. Thus the estimated standard error is 3.0. Using $t^* = 2.06$, which 95% confidence interval is obtained?

- A) (4.20, 10.20)
- B) (2.26, 12.14)
- C) (1.02, 13.38)
- D) (-1.44, 15.84)

APSTAT-MCQ10-010

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

An infographic represents a category with frequency 40 by using an icon twice as tall and twice as wide as the icon for a category with frequency 20. How does the icon area compare with the actual frequency ratio?

- A) The icon areas have the correct 2-to-1 ratio.
- B) The icon for frequency 40 has only 1.5 times the area.
- C) The icon areas are equal because the same shape is used.
- D) The icon for frequency 40 has 4 times the area, exaggerating the actual 2-to-1 ratio.

APSTAT-MCQ10-011

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

Two one-predictor regressions use the same observations and response, solar-panel output. Model A has $r^2 = 0.49$ and residual standard deviation 12 watt-hours; model B has $r^2 = 0.64$ and residual standard deviation 10 watt-hours. Which comparison is correct?

- A) Model B accounts for more response variation and has smaller typical vertical prediction errors on these data.
- B) Model A is better because its r^2 is closer to its residual standard deviation.
- C) Model B proves that its explanatory variable causes output to change.
- D) Model A must have the stronger correlation because 12 is greater than 10.

APSTAT-MCQ10-012

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Formulate Questions

Prompt:

A wildlife team wants to test whether the population proportion of tagged birds that return differs from the population proportion of untagged birds that return. Which hypotheses are appropriate?

- A) $H_0: p_{\text{tagged}} - p_{\text{untagged}} = 0$; $H_a: p_{\text{tagged}} - p_{\text{untagged}} \neq 0$
- B) $H_0: \hat{p}_{\text{tagged}} - \hat{p}_{\text{untagged}} = 0$; $H_a: \hat{p}_{\text{tagged}} - \hat{p}_{\text{untagged}} > 0$
- C) $H_0: p_{\text{tagged}} = 0$; $H_a: p_{\text{untagged}} = 0$
- D) $H_0: \mu_{\text{tagged}} - \mu_{\text{untagged}} = 0$; $H_a: \mu_{\text{tagged}} - \mu_{\text{untagged}} \neq 0$

APSTAT-MCQ10-013

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Collect Data

Prompt:

In a randomized fertilizer experiment, researchers observe a difference in mean plant growth between two treatment groups. Which simulation best approximates the null randomization distribution?

- A) Generate new growth values from a normal distribution and keep the original treatment labels.
- B) Resample only plants from the treatment group with the larger observed mean.
- C) Repeatedly shuffle the existing treatment labels among all observed growth values and record the difference in group means each time.
- D) Repeatedly add the observed difference to both group means and record the total growth.

APSTAT-MCQ10-014

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Interpret Results

Prompt:

In a strongly left-skewed distribution, one observation has z-score -2 . Which conclusion is justified?

- A) The observation is 2 standard deviations below the mean, but a normal table cannot be used to assign it the 2.5th percentile without a normal model.
- B) The observation must be below exactly 2.5% of the data.
- C) The observation is 2 raw units below the median.
- D) The observation must be a potential outlier by the $1.5 \times \text{IQR}$ rule.

APSTAT-MCQ10-015

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

For 25 paired observations, the differences are defined as new minus current, with $\bar{x}_d = -1.8$ and $s_d = 4.0$. Using $t^* = 1.711$, what is the 95% one-sided upper confidence bound for μ_d , and what does it support?

- A) 0.60; it supports $\mu_d > 0$
- B) -5.22 ; it supports $\mu_d < -5$
- C) 1.62; it does not distinguish the methods
- D) Approximately -0.43 ; it supports a negative population mean difference

APSTAT-MCQ10-016

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Collect Data

Prompt:

Eligible users are randomly assigned to two account-verification interfaces, and completion proportions will be compared. Outcome data are missing for 5% of users assigned to interface A and 35% assigned to interface B. Which assessment is best?

- A) Random assignment guarantees an unbiased comparison regardless of missing outcomes.
- B) The missing outcomes matter only if both groups have exactly the same completion proportion.
- C) The study should automatically count every missing outcome as a completion.

D) Differential attrition can destroy comparability of the observed groups and threaten the causal conclusion.

APSTAT-MCQ10-017

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Formulate Questions

Prompt:

Which research question most clearly calls for a randomized experiment comparing means?

- A) Among eligible office workers, how does mean uninterrupted focus time differ when workers are randomly assigned to receive either bundled or immediate notifications?
- B) What proportion of all office workers own a smartphone?
- C) Is department name associated with preferred notification color?
- D) What is the median number of messages sent by one selected worker yesterday?

Questions APSTAT-MCQ10-018 to APSTAT-MCQ10-020 refer to the following stimulus.

The settling time X , in seconds, of a microscopic droplet has cumulative distribution function $F(x) = \sqrt{x/9}$ for $0 \leq x \leq 9$, with $F(x) = 0$ below 0 and $F(x) = 1$ above 9.

APSTAT-MCQ10-018

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

For a droplet settling time with $F(x) = \sqrt{x/9}$ on $0 \leq x \leq 9$, what is $P(X \leq 1 \text{ second})$?

- A) $1/9$
- B) $1/3$
- C) $2/3$
- D) $8/9$

APSTAT-MCQ10-019

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

For a droplet settling time with $F(x) = \sqrt{x/9}$ on $0 \leq x \leq 9$, what is $P(X > 4 \mid X > 1)$?

- A) $1/3$
- B) $2/3$
- C) $1/2$
- D) $3/4$

APSTAT-MCQ10-020

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

For a droplet settling time with $F(x) = \sqrt{x/9}$ on $0 \leq x \leq 9$, which value is the median and what does it mean?

- A) 2.25 seconds; half of settling times are at or below 2.25 seconds
- B) 4.50 seconds; the support midpoint is always the median
- C) 6.75 seconds; 75% of settling times are below it
- D) 3.00 seconds; the median equals $\sqrt{9}$

APSTAT-MCQ10-021

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

A least-squares line is $\hat{y} = b_0 + b_1x$. The explanatory variable is recoded as $x' = x/10$, while y is unchanged. Which statement is correct for the line written in terms of x' ?

- A) The slope becomes $10b_1$, while the intercept, fitted values, residuals, and correlation remain unchanged.
- B) The slope becomes $b_1/10$ and the correlation becomes $r/10$.
- C) The intercept becomes $10b_0$ and the residuals are divided by 10.
- D) Both slope and intercept remain unchanged because x and x' contain the same information.

APSTAT-MCQ10-022

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Collect Data

Prompt:

A licensing board wants to estimate the proportion of all licensed professionals who completed new ethics training. Completion rates may differ by specialty, and the roster gives each specialty's population size. Which sampling plan is best?

- A) Take the same number of volunteers from every specialty and report the unweighted overall proportion.
- B) Stratify the roster by specialty and take independent random samples allocated in proportion to specialty population sizes.
- C) Sample only the largest specialty because it contains the most professionals.
- D) Select one specialty at random and treat its members as an SRS of all professionals.

APSTAT-MCQ10-023

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

For the four observations 2, 5, 9, and 20, define $L(c) = |2 - c| + |5 - c| + |9 - c| + |20 - c|$. Which choice of c minimizes $L(c)$?

- A) Only $c = 2$
- B) Only $c = 7$
- C) Any value of c from 5 through 9, inclusive
- D) Only $c = 20$

APSTAT-MCQ10-024

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Formulate Questions

Prompt:

Each of 18 farm blocks is divided into two similar plots. Within every block, one plot is randomly assigned fertilizer A and the other fertilizer B. Which procedure is appropriate for estimating the population mean yield difference A minus B?

- A) A one-proportion z-interval based on which fertilizer wins each block
- B) An independent two-sample t-interval that ignores the blocks
- C) A paired t-interval applied to the 18 within-block yield differences
- D) A one-sample t-interval using only the fertilizer A yields

APSTAT-MCQ10-025

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Three independent requests are each assigned uniformly to one of four servers. What is the probability that exactly two requests go to the same server and the third goes to a different server?

- A) $9/16$
- B) $3/16$
- C) $3/8$
- D) $13/16$

APSTAT-MCQ10-026

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A randomization test uses 999 shuffled treatment assignments. Fourteen shuffled assignments produce a difference in sample proportions at least as extreme as the observed difference. Using the plus-one calculation that includes the observed assignment, what is the estimated p-value?

- A) 0.014
- B) 0.015
- C) 0.140
- D) 0.150

APSTAT-MCQ10-027

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A campus wants to estimate the proportion of waste placed in the correct recycling bin. Visible observers may change people's behavior. Which plan best reduces this measurement effect while retaining broad coverage?

- A) Ask environmentally active volunteers how accurately everyone recycles.
- B) Observe only the bin beside the sustainability office during one lunch period.

- C) Randomly select location-time windows and conduct unobtrusive coded audits of deposited items using one protocol.
- D) Announce each observation period and interview only people who approach the observer.

APSTAT-MCQ10-028

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

A one-sample t statistic is positive, and its two-sided p -value is 0.08. What is the p -value for the left-tailed alternative $H_a: \mu < \mu_0$?

- A) 0.04
- B) 0.08
- C) 0.92
- D) 0.96

APSTAT-MCQ10-029

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Interpret Results

Prompt:

Independent SRSs of remote and onsite employees from a company show a statistically significant difference in mean weekly meeting time. Work arrangement was not randomly assigned. Which conclusion is justified?

- A) Remote work causes the observed difference for every company employee.
- B) The result cannot be generalized because random assignment was absent.
- C) The two population means must differ by a practically important amount.
- D) The result supports a difference in company-population means, but it does not establish that work arrangement caused the difference.

APSTAT-MCQ10-030

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A dataset of 12 observations can be arranged into 6 pairs so that the two values in every pair sum to 100, with the pairing symmetric from the smallest and largest values inward. What must be true?

- A) The mean is 100 and the median cannot be determined.
- B) Both the mean and the median are 50.
- C) The mean is 50, but the median must be one of the observed values.
- D) The standard deviation is 50.

APSTAT-MCQ10-031

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Formulate Questions

Prompt:

A researcher wants to assess whether the sample median is biased for a skewed population when $n = 9$. Which simulation plan directly addresses that goal?

- A) Generate one sample of size 9 and compare its mean with its median.
- B) Generate many random samples of size 9, save each sample median, and compare the mean of those medians with the population median.
- C) Generate many samples of different sizes and save only their maximum observations.
- D) Repeatedly record the fixed population median without generating samples.

APSTAT-MCQ10-032

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

A monitoring agency considers failing to detect a real increase in a contamination proportion much more costly than a false alarm. With sample size fixed, which change is consistent with that priority?

- A) Lower α , which lowers both Type I and Type II error probabilities.
- B) Lower α , which raises power against every alternative.
- C) Keep α fixed because the relative costs cannot inform a test design.
- D) Use a larger α to reduce Type II error and increase power, accepting a higher Type I error rate.

APSTAT-MCQ10-033

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

Fruit masses from orchard A are approximately normal with mean 50 grams and standard deviation 5 grams. Fruit masses from orchard B are approximately normal with mean 55 grams and standard deviation 10 grams. Use $\Phi(2.00) = 0.9772$ and $\Phi(0.50) = 0.6915$. Which orchard has the larger proportion of fruit above 60 grams?

- A) Orchard A, because 60 is closer to 50 than to 55 in raw units
- B) The proportions are equal because 60 exceeds both means
- C) Orchard B, because its upper-tail probability is $1 - 0.6915 = 0.3085$, compared with $1 - 0.9772 = 0.0228$ for orchard A
- D) Orchard A, because its standard deviation is smaller

APSTAT-MCQ10-034

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

In a paired study, 3 units are added to both the before and after measurement for every participant to correct a calibration offset. What happens to a paired t interval and paired t test based on after-minus-before differences?

- A) The interval center increases by 6 and the t statistic doubles.
- B) The interval center increases by 3 but the p -value is unchanged.
- C) The standard error increases because every measurement is larger.
- D) The differences, interval, t statistic, and p -value all remain unchanged.

Questions APSTAT-MCQ10-035 to APSTAT-MCQ10-037 refer to the following stimulus.

For 15 load tests, a regression predicts response latency y , in milliseconds, from concurrent sessions x , in thousands. With all observations, $\hat{y} = 5 + 0.9x$ and $r = 0.82$. The point P is $(30, 32)$. Without P , the line is $\hat{y} = 12 + 0.2x$ and $r = 0.31$.

APSTAT-MCQ10-035

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

For the latency regression, the line changes from $\hat{y} = 5 + 0.9x$ with $r = 0.82$ to $\hat{y} = 12 + 0.2x$ with $r = 0.31$ when $P = (30, 32)$ is removed. What does this show about P ?

- A) P is not influential because its observed y -value is an integer.

- B) P is an outlier only in the y-direction and has no effect on association.
- C) P is influential because removing it substantially changes both the fitted line and correlation.
- D) P proves that concurrent sessions cause higher latency.

APSTAT-MCQ10-036

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

Using the all-observation latency line $\hat{y} = 5 + 0.9x$, what is the residual for $P = (30, 32)$?

- A) -27 milliseconds
- B) -5 milliseconds
- C) 0 milliseconds
- D) 32 milliseconds

APSTAT-MCQ10-037

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

Point $P = (30, 32)$ has residual 0 under the all-observation line $\hat{y} = 5 + 0.9x$, yet removing it changes the line to $\hat{y} = 12 + 0.2x$. Which explanation is best?

- A) A zero residual guarantees that a point cannot influence a regression.
- B) P's extreme x-value gives it high leverage, allowing it to pull the fitted line toward itself even though its final residual is 0.
- C) Residual 0 means P must be located at $(\bar{x}, \bar{y})$.
- D) Correlation is determined only by residual size, not by x-position.

APSTAT-MCQ10-038

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

A classifier's odds in favor of a correct result are 3:2 for each image. Results for five images are independent. What is the probability of at least one incorrect result among the five images?

- A) $1 - 0.6^5 = 0.92224$
- B) $0.4^5 = 0.01024$
- C) $5(0.6) = 3.0$
- D) $1 - 0.4^5 = 0.98976$

APSTAT-MCQ10-039

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A planning calculation shows that 385 completed responses are required for a proportion interval. If the anticipated response rate is 80%, what minimum number of randomly selected people should initially be invited so that the expected number of completed responses is at least 385?

- A) 385
- B) 482
- C) 308
- D) 770

APSTAT-MCQ10-040

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

A two-sided test of $H_0: p = 0.50$ fails to reject. Before data collection, the design had power 0.95 when $p = 0.60$. Which interpretation is best?

- A) The result proves $p = 0.50$ because a high-power study cannot make an error.
- B) The probability that $p = 0.60$ is exactly 0.05.
- C) The result establishes that every alternative proportion is false.
- D) The result does not prove $p = 0.50$, but a failure to reject would be unusual if p were 0.60, so that specific large departure is less compatible with the data and design.

APSTAT-MCQ10-041

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A city wants a probability sample of adult residents for a transportation survey. Which plan provides the strongest coverage and within-household selection?

- A) Call numbers listed in a landline directory and interview whoever answers.
- B) Place a survey link on the city's social-media account.
- C) Randomly select residential addresses from a complete address frame, then randomly select one adult within each responding household and offer multiple response modes.
- D) Interview adults waiting at the central transit station.

APSTAT-MCQ10-042

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

An agricultural service compares growers' water use after launching a new irrigation advisory with water use among last year's growers. Which redesign best separates the advisory effect from year-to-year changes?

- A) Use only growers who report that they liked the advisory.
- B) During the same season, randomly assign eligible growers to receive the advisory or the usual information and compare water use.
- C) Increase the number of last-year records without creating a concurrent comparison group.
- D) Compare this year's advisory users with this year's growers who voluntarily declined it.

PART B, ANSWER KEY AND EXPLANATIONS

APSTAT-MCQ10-001, Correct: A, Correct Answer: Mean = 1.875 and median = 1.5, Explanation: The total is $4(1) + 3(2) + 1(5) = 15$ across 8 visits, so the mean is $15/8 = 1.875$. The fourth ordered value is 1 and the fifth is 2, so the median is $(1 + 2)/2 = 1.5$.

APSTAT-MCQ10-002, Correct: D, Correct Answer: Overlap of two separate intervals does not by itself determine the two-proportion test; the standard error of $\hat{p}_A - \hat{p}_B$ must be analyzed directly., Explanation: Separate intervals contain uncertainty for two individual estimates, whereas the test concerns the sampling distribution of their difference. Slight overlap can coexist with a significant difference, so the dedicated two-proportion procedure is required.

APSTAT-MCQ10-003, Correct: B, Correct Answer: 0.30, Explanation: Since $P(A \cup B)$ cannot exceed 1, $P(A \cap B) = P(A) + P(B) - P(A \cup B)$ is at least $0.70 + 0.60 - 1 = 0.30$. This lower bound is attainable when $A \cup B$ is the entire sample space.

APSTAT-MCQ10-004, Correct: A, Correct Answer: $\mu_A - \mu_B$, Explanation: The target concerns two population mean fermentation times, in the stated A-minus-B order. The difference of sample means is a statistic, not the parameter, and the batches are independent rather than paired.

APSTAT-MCQ10-005, Correct: A, Correct Answer: It increases squared residual variation and weakens $|r|$, but its x-position gives it relatively little leverage on the slope., Explanation: The point is vertically unusual, so it adds a large residual and weakens linear association. Leverage is driven by distance in the x-direction, and an x-value near $\bar{x}$ has relatively low leverage.

APSTAT-MCQ10-006, Correct: C, Correct Answer: Draw an SRS from the membership roster, contact selected members repeatedly through several modes, and document response patterns., Explanation: The SRS supplies known selection probabilities. Repeated multimode follow-up can improve response and documenting patterns permits assessment of remaining nonresponse; volunteers and convenience replacements do not preserve the original probability design.

APSTAT-MCQ10-007, Correct: B, Correct Answer: $z = 0$ and $p\text{-value} = 1$, Explanation: Both sample proportions equal 0.60, so the observed difference is 0. The pooled standard error is positive, giving $z = 0/SE = 0$; a two-sided statistic of zero has $p\text{-value} = 1$.

APSTAT-MCQ10-008, Correct: D, Correct Answer: Rain and lateness are independent in the probability model because conditioning on rain does not change the late-delivery probability., Explanation: For an event with positive probability, $P(\text{late} \mid \text{rain}) = P(\text{late})$ is a definition of independence. It does not establish causation or require equal daily counts.

APSTAT-MCQ10-009, Correct: C, Correct Answer: (1.02, 13.38), Explanation: The margin of error is $t^*(SE) = 2.06(3.0) = 6.18$. Therefore the interval is 7.2 ± 6.18 , or (1.02, 13.38).

APSTAT-MCQ10-010, Correct: D, Correct Answer: The icon for frequency 40 has 4 times the area, exaggerating the actual 2-to-1 ratio., Explanation: Doubling both height and width multiplies area by $2^2 = 4$. The underlying frequency only doubles, so the area-based visual impression exaggerates the comparison.

APSTAT-MCQ10-011, Correct: A, Correct Answer: Model B accounts for more response variation and has smaller typical vertical prediction errors on these data., Explanation: Model B has the larger r^2 , so it accounts for a larger fraction of observed response variation. Its smaller residual standard deviation indicates smaller typical vertical errors; neither comparison establishes causation.

APSTAT-MCQ10-012, Correct: A, Correct Answer: $H_0: p_{\text{tagged}} - p_{\text{untagged}} = 0$; $H_a: p_{\text{tagged}} - p_{\text{untagged}} \neq 0$, Explanation: The question concerns a difference between two population proportions and uses nondirectional wording, so the alternative is two-sided. Hypotheses must use population parameters rather than sample statistics.

APSTAT-MCQ10-013, Correct: C, Correct Answer: Repeatedly shuffle the existing treatment labels among all observed growth values and record the difference in group means each time., Explanation: Under a no-treatment-effect null, the labels are exchangeable because they were randomly assigned. Relabeling preserves the observed outcomes and recreates differences attributable to assignment alone.

APSTAT-MCQ10-014, Correct: A, Correct Answer: The observation is 2 standard deviations below the mean, but a normal table cannot be used to assign it the 2.5th percentile without a normal model., Explanation: A z-score standardizes distance from the mean regardless of distribution shape. Normal tail areas require an approximately normal model, which is contradicted by the stated strong skew.

APSTAT-MCQ10-015, Correct: D, Correct Answer: Approximately -0.43 ; it supports a negative population mean difference, Explanation: $SE = 4/\sqrt{25} = 0.8$. The upper bound is $-1.8 + 1.711(0.8) = -0.4312$, approximately -0.43 . Because even the upper bound is below 0, the data support $\mu_d < 0$.

APSTAT-MCQ10-016, Correct: D, Correct Answer: Differential attrition can destroy comparability of the observed groups and threaten the causal conclusion., Explanation: Random assignment makes the assigned groups comparable initially, but substantially different missing-outcome rates can select different kinds of observed users in the two groups. Randomization does not automatically repair that postassignment loss.

APSTAT-MCQ10-017, Correct: A, Correct Answer: Among eligible office workers, how does mean uninterrupted focus time differ when workers are randomly assigned to receive either bundled or immediate notifications?, Explanation: The response is quantitative, two treatment groups are compared by their means, and random assignment makes this an experimental causal question. The other choices ask about one proportion, categorical association, or one person's value.

APSTAT-MCQ10-018, Correct: B, Correct Answer: $1/3$, Explanation: By definition of the cumulative distribution function, $P(X \leq 1) = F(1) = \sqrt{(1/9)} = 1/3$.

APSTAT-MCQ10-019, Correct: C, Correct Answer: $1/2$, Explanation: $P(X > 4) = 1 - F(4) = 1 - 2/3 = 1/3$, and $P(X > 1) = 1 - 1/3 = 2/3$. Because $X > 4$ is contained in $X > 1$, the conditional probability is $(1/3)/(2/3) = 1/2$.

APSTAT-MCQ10-020, Correct: A, Correct Answer: 2.25 seconds; half of settling times are at or below 2.25 seconds, Explanation: The median m satisfies $F(m) = 0.50$. Solving $\sqrt{(m/9)} = 0.50$ gives $m/9 = 0.25$ and $m = 2.25$ seconds.

APSTAT-MCQ10-021, Correct: A, Correct Answer: The slope becomes $10b_1$, while the intercept, fitted values, residuals, and correlation remain unchanged., Explanation: Since $x = 10x'$, substitution gives $\hat{y} = b_0 + b_1(10x') = b_0 + 10b_1x'$. This recoding describes the same points and predictions, so residuals and correlation do not change.

APSTAT-MCQ10-022, Correct: B, Correct Answer: Stratify the roster by specialty and take independent random samples allocated in proportion to specialty population sizes., Explanation: Stratification ensures representation of every specialty, while proportional allocation makes the combined unweighted estimate align with the population composition. Volunteers or one selected specialty do not represent the full roster.

APSTAT-MCQ10-023, Correct: C, Correct Answer: Any value of c from 5 through 9, inclusive, Explanation: A sum of absolute deviations is minimized at a median. With an even number of observations, every value between the two middle ordered observations, 5 and 9, gives the same minimum total absolute deviation.

APSTAT-MCQ10-024, Correct: C, Correct Answer: A paired t-interval applied to the 18 within-block yield differences, Explanation: Each block supplies a natural pair because both treatments appear on similar plots within the block. One A-minus-B difference per block removes much of the block-to-block variation.

APSTAT-MCQ10-025, Correct: A, Correct Answer: 9/16, Explanation: Among $4^3 = 64$ equally likely server assignments, choose the repeated server in 4 ways, the different server in 3 ways, and the position of the different request in 3 ways. This gives $4(3)(3) = 36$ outcomes, so the probability is $36/64 = 9/16$.

APSTAT-MCQ10-026, Correct: B, Correct Answer: 0.015, Explanation: The plus-one estimate is $(14 + 1)/(999 + 1) = 15/1000 = 0.015$. The adjustment includes the observed assignment among the possible randomization outcomes.

APSTAT-MCQ10-027, Correct: C, Correct Answer: Randomly select location-time windows and conduct unobtrusive coded audits of deposited items using one protocol., Explanation: Random location-time coverage supports representation, and unobtrusive standardized measurement reduces behavior changes caused by visible observation. Convenience locations and volunteers can be systematically unrepresentative.

APSTAT-MCQ10-028, Correct: D, Correct Answer: 0.96, Explanation: With a positive statistic, the upper-tail area is half the two-sided p-value, $0.08/2 = 0.04$. The left-tail area for the opposite alternative is $1 - 0.04 = 0.96$.

APSTAT-MCQ10-029, Correct: D, Correct Answer: The result supports a difference in company-population means, but it does not establish that work arrangement caused the difference., Explanation: Independent SRSs support generalization to the two company employee populations. Because work arrangement was not randomly assigned, lurking variables may explain the observed mean difference, so causation is not justified.

APSTAT-MCQ10-030, Correct: B, Correct Answer: Both the mean and the median are 50., Explanation: The six pair sums total $6(100) = 600$, so the mean is $600/12 = 50$. Symmetric inward pairing makes the two middle ordered values sum to 100, so their average, the median, is also 50.

APSTAT-MCQ10-031, Correct: B, Correct Answer: Generate many random samples of size 9, save each sample median, and compare the mean of those medians with the population median., Explanation: Bias concerns the center of an estimator's repeated-sampling distribution relative to its population target. Repeated sample medians approximate that distribution, and comparing their average with the population median assesses bias.

APSTAT-MCQ10-032, Correct: D, Correct Answer: Use a larger α to reduce Type II error and increase power, accepting a higher Type I error rate., Explanation: With n fixed, enlarging the rejection region by increasing α makes rejection more likely under both the null and alternatives. This raises Type I error but reduces β and increases power, matching the stated cost priority.

APSTAT-MCQ10-033, Correct: C, Correct Answer: Orchard B, because its upper-tail probability is $1 - 0.6915 = 0.3085$, compared with $1 - 0.9772 = 0.0228$ for orchard A, Explanation: For orchard A, $z = (60 - 50)/5 = 2$, giving upper tail 0.0228. For orchard B, $z = (60 - 55)/10 = 0.5$, giving upper tail 0.3085, so orchard B has the larger proportion above 60 grams.

APSTAT-MCQ10-034, Correct: D, Correct Answer: The differences, interval, t statistic, and p-value all remain unchanged., Explanation: For every participant, $(\text{after} + 3) - (\text{before} + 3) = \text{after} - \text{before}$. Because all paired differences are unchanged, every summary and inference computed from them is unchanged.

APSTAT-MCQ10-035, Correct: C, Correct Answer: P is influential because removing it substantially changes both the fitted line and correlation., Explanation: Influence is assessed by how much a fitted analysis changes when a point is removed. The slope changes from 0.9 to 0.2 and r from 0.82 to 0.31, showing substantial influence.

APSTAT-MCQ10-036, Correct: C, Correct Answer: 0 milliseconds, Explanation: At $x = 30$, the fitted latency is $5 + 0.9(30) = 32$ milliseconds. Residual = observed – predicted = $32 - 32 = 0$ milliseconds.

APSTAT-MCQ10-037, Correct: B, Correct Answer: P's extreme x -value gives it high leverage, allowing it to pull the fitted line toward itself even though its final residual is 0., Explanation: A point far from the other x -values can strongly affect the least-squares line. The fitted line can be pulled through such a point, leaving a small residual that conceals its influence unless the fit is recomputed without it.

APSTAT-MCQ10-038, Correct: A, Correct Answer: $1 - 0.6^5 = 0.92224$, Explanation: Odds 3:2 correspond to correct-result probability $3/(3 + 2) = 0.6$. The probability all five are correct is $0.6^5 = 0.07776$, so the complement, at least one incorrect, is 0.92224.

APSTAT-MCQ10-039, Correct: B, Correct Answer: 482, Explanation: If N people are invited, the expected number of responses is $0.80N$. Requiring $0.80N \geq 385$ gives $N \geq 481.25$, so the initial invitation count must be rounded up to 482.

APSTAT-MCQ10-040, Correct: D, Correct Answer: The result does not prove $p = 0.50$, but a failure to reject would be unusual if p were 0.60, so that specific large departure is less compatible with the data and design., Explanation: Power 0.95 at $p = 0.60$ means the probability of failing to reject there is 0.05. Nonrejection never proves the null, but high power against a specified alternative makes that particular alternative less consistent with the outcome.

APSTAT-MCQ10-041, Correct: C, Correct Answer: Randomly select residential addresses from a complete address frame, then randomly select one adult within each responding household and offer multiple response modes., Explanation: Address-based sampling covers households regardless of telephone listing, and random within-household selection avoids interviewing only the easiest adult to reach. Multiple response modes can improve access without replacing probability selection.

APSTAT-MCQ10-042, Correct: B, Correct Answer: During the same season, randomly assign eligible growers to receive the advisory or the usual information and compare water use., Explanation: A concurrent randomized control group experiences the same season and separates the advisory from year effects. Historical comparisons and voluntary uptake can confound the advisory with time or self-selection.

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-MCQ10

· APSTAT-MCQ10-001 to 005: Unit 1 a return-count frequency table with values 1, 2, and 5 jointly calculates mean 1.875 and median 1.5; Unit 3 slightly overlapping separate variety-proportion intervals do not determine a test for $p_A - p_B$; Unit 2 marginals 0.70 and 0.60 force a minimum intersection of 0.30 because a union cannot exceed 1; Unit 4 independent recipe-batch fermentation times target $\mu_A - \mu_B$; Unit 5 a vertical outlier near $\bar{x}$ increases residual variation while having low leverage. Misconceptions include ignoring frequencies or middle positions, treating separate-interval overlap as a formal difference test, allowing impossible event overlap, using a statistic as a parameter, and equating vertical unusualness with high leverage.

- APSTAT-MCQ10-006 to 010: Unit 1 an association membership SRS uses repeated multimode contact rather than volunteer replacement; Unit 3 equal sample proportions 54/90 and 30/50 yield $z = 0$ and two-sided $p = 1$; Unit 2 equality of $P(\text{late} \mid \text{rain})$ and $P(\text{late})$ establishes model independence; Unit 4 a reported estimate 7.2 and $t = 2.40$ recover $SE = 3$ before constructing (1.02, 13.38); Unit 1 an icon doubled in both dimensions quadruples area and exaggerates a twofold frequency. Misconceptions include abandoning probability selection after nonresponse, assuming unequal n forces a nonzero z , confusing independence with causation, using t rather than t^* in an interval, and scaling pictograms by two dimensions.
- APSTAT-MCQ10-011 to 015: Unit 5 two solar-output models on the same observations compare r^2 values 0.49 versus 0.64 and residual standard deviations 12 versus 10 watt-hours; Unit 3 tagged-versus-untagged bird-return proportions use a two-sided population-parameter hypothesis; Unit 2 a fertilizer randomization distribution shuffles treatment labels over fixed growth outcomes; Unit 1 $z = -2$ in a strongly skewed distribution cannot receive a normal percentile; Unit 4 paired new-minus-current differences construct a one-sided upper t bound -0.43 . Misconceptions include preferring a larger residual spread or claiming causation, putting sample statistics in hypotheses, generating new outcomes instead of relabeling, using normal tables without a normal model, and forgetting $s_d/\sqrt{n}$.
- APSTAT-MCQ10-016 to 020: Unit 3 differential 5% versus 35% interface attrition threatens postrandomization comparability; Unit 1 a notification experiment formulates a randomized comparison of mean focus times; Unit 2 a droplet settling-time CDF $F(x) = \sqrt{x/9}$ set calculates $F(1) = 1/3$, conditional tail probability $1/2$, and median 2.25 seconds. Misconceptions include assuming randomization repairs missing outcomes, selecting a proportion or one-person question instead of a mean experiment, confusing a CDF with a density, omitting conditional restriction, and using the support midpoint as the median.
- APSTAT-MCQ10-021 to 025: Unit 5 recoding $x' = x/10$ multiplies regression slope by 10 while preserving predictions and r ; Unit 3 proportional stratification by professional specialty represents the licensing roster; Unit 1 observations 2, 5, 9, and 20 minimize total absolute deviation for every center c between the two middle values 5 and 9; Unit 4 paired farm plots within 18 blocks use a t interval on yield differences; Unit 2 three requests assigned to four servers count exactly-one-pair patterns as $36/64 = 9/16$. Misconceptions include scaling r with units, treating equal subgroup samples as self-weighting, assuming only the midpoint 7 can minimize absolute distance, ignoring blocks, and omitting the choice or position of the singleton server.
- APSTAT-MCQ10-026 to 030: Unit 3 a 999-shuffle randomization test uses the plus-one p -value $15/1000$; Unit 1 random location-time recycling audits reduce observer effects; Unit 4 positive t with two-sided $p = 0.08$ gives opposite left-tail $p = 0.96$; Unit 4 remote-versus-onsite employee SRSs support population comparison but not causation; Unit 1 twelve values symmetrically paired to sum 100 force mean and median 50. Misconceptions include dividing 14 by 999 without the supplied adjustment, using visible convenience observation, halving the wrong tail, treating random sampling as assignment, and overlooking symmetry of the two middle values.
- APSTAT-MCQ10-031 to 035: Unit 2 repeated size-9 samples save medians and compare their sampling-distribution mean with the population median to assess estimator bias; Unit 3 larger α trades more false alarms for lower β when missed contamination is costlier; Unit 1 two normal orchard distributions compare upper tails at $z = 2$ and $z = 0.5$; Unit 4 adding the same calibration constant to both paired measurements leaves all differences and t inference unchanged; Unit 5 a latency-regression stimulus identifies $P = (30, 32)$ as influential from large changes in line and r . Misconceptions include comparing one sample's mean and median rather than repeated medians with the target, expecting both error rates to fall, comparing raw cutoffs without standardizing, shifting paired differences, and judging influence only by a point's residual.
- APSTAT-MCQ10-036 to 040: Unit 5 the latency stimulus calculates P 's zero residual and explains high-leverage influence despite that residual; Unit 2 odds 3:2 become $p = 0.6$ and five independent

classifications give at-least-one-error probability $1 - 0.6^5$; Unit 3 a requirement of 385 completed proportion responses with 80% anticipated response inflates the initial invitation count to 482; Unit 3 nonrejection in a test with power 0.95 at $p = 0.60$ weakens that specified large alternative without proving $p = 0.50$. Misconceptions include equating zero residual with no influence, using 0.4^5 rather than the all-correct complement, multiplying rather than dividing by the response rate or rounding down, and interpreting nonrejection as proof.

- APSTAT-MCQ10-041 to 042: Unit 1 an address-based adult survey randomly selects one adult per sampled household and offers multiple modes; Unit 1 a same-season randomized irrigation-advisory experiment replaces a historical comparison. Misconceptions include landline-only, social-media, or transit-station frames, and confounding an advisory with year effects or voluntary uptake.

- Unit counts for APSTAT-MCQ10: Unit 1 = 11; Unit 2 = 9; Unit 3 = 9; Unit 4 = 7; Unit 5 = 6. Skill counts: Formulate Questions = 5; Collect Data = 7; Analyze Data = 21; Interpret Results = 9. Stimulus sets: continuous droplet settling-time CDF, APSTAT-MCQ10-018 to 020; influential high-leverage latency observation, APSTAT-MCQ10-035 to 037.